

BORYSPIL, Ukraine

WMO: 333470

Lat: 50.3339N

Lon: 30.9492E

Elev: 400

StdP: 14.48

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.0	(c) 2.9	(d) -7.9	(e) 3.6	(f) -3.1	(g) -2.1	(h) 5.0	(i) 3.4	(j) 23.0	(k) 25.5	(l) 21.1	(m) 25.4	(n) 6.0	(o) 20	(p) 0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.6	(c) 89.4	(d) 68.1	(e) 86.0	(f) 67.3	(g) 82.7	(h) 65.7	(i) 70.9	(j) 82.7	(k) 69.5	(l) 81.0	(m) 68.1	(n) 79.2	(o) 9.0	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.7	103.2	75.5	66.0	97.4	73.8	64.4	91.9	72.4	35.1	83.1	33.8	80.9	32.7	79.3	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.5	(b) 18.2	(c) 16.3	(e) -10.6	(f) 93.8	(g) 7.0	(h) 3.8	(i) -15.6	(j) 96.5	(k) -19.7	(l) 98.8	(m) -23.6	(n) 100.9	(o) -28.7	(p) 103.7		
(4) 20.5	(5) 18.2	(6) 16.3	(7) -11.3	(8) 74.2	(9) 7.1	(10) 1.4	(11) -16.5	(12) 75.3	(13) -20.6	(14) 76.1	(15) -24.6	(16) 76.9	(17) -29.7	(18) 78.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	24.1	26.1	35.0	48.7	59.4	65.4	69.1	67.1	57.5	45.9	36.4	27.7		
	DBStd	18.00	10.57	10.95	8.34	7.47	7.57	6.14	5.32	6.04	6.95	7.92	8.84	10.06		
	HDD50	3356	803	670	466	111	13	0	0	0	14	174	414	691		
	HDD65	6968	1268	1090	930	489	205	71	21	46	242	591	859	1156		
	CDD50	2250	0	0	1	73	303	462	591	530	238	48	4	0		
	CDD65	387	0	0	0	1	30	83	147	110	16	0	0	0		
	CDH74	4437	0	0	0	23	393	920	1532	1318	243	8	0	0		
	CDH80	1396	0	0	0	4	83	256	514	485	54	0	0	0		
	WSAvg	8.2	9.0	9.2	9.4	8.6	8.1	7.8	7.2	7.0	7.4	7.7	8.6	8.8		
	PrecAvg	22.0	1.2	1.3	1.3	1.4	2.2	2.8	2.7	2.2	2.1	1.6	1.6	1.5		
(15) Precipitation	PrecMax	27.8	2.2	3.0	4.2	4.3	6.3	6.2	6.1	4.6	7.4	4.2	3.3	4.8		
	PrecMin	16.4	0.2	0.2	0.1	0.0	0.7	0.2	0.7	0.1	0.0	0.0	0.0	0.3		
	PrecStd	2.7	0.6	0.6	0.9	0.9	1.3	1.5	1.3	1.2	1.9	1.0	0.8	1.0		
	PrecStd	2.7	0.6	0.6	0.9	0.9	1.3	1.5	1.3	1.2	1.9	1.0	0.8	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.9	48.6	61.1	77.1	86.4	89.6	92.9	95.3	85.9	73.8	60.5	49.9		
		MCWB	41.3	43.7	48.6	59.3	65.1	67.8	69.4	69.0	64.0	58.9	54.3	47.8		
	2%	DB	41.2	44.5	55.7	71.3	82.2	85.9	89.2	89.7	80.5	68.2	55.0	44.9		
		MCWB	39.2	41.2	45.8	54.7	62.5	66.5	68.5	68.2	62.5	57.5	51.6	43.4		
	5%	DB	38.8	40.9	51.6	66.5	78.6	82.7	85.9	85.6	75.6	62.6	50.4	42.6		
		MCWB	37.1	38.6	44.1	53.2	62.1	65.7	68.3	67.1	60.9	55.4	48.2	41.6		
	10%	DB	35.8	37.6	46.8	62.8	74.9	79.2	82.4	80.9	71.4	58.6	48.2	39.2		
		MCWB	34.6	35.8	41.8	51.4	60.9	64.8	67.0	65.2	59.3	52.8	46.6	38.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.4	45.1	50.6	60.3	68.5	71.9	73.3	72.6	67.3	62.7	55.2	48.1		
		MCDB	44.0	47.8	59.5	72.8	80.3	82.7	84.4	85.7	80.4	71.5	58.8	49.2		
	2%	WB	39.3	41.1	47.4	56.9	65.9	69.6	71.4	70.2	64.4	59.2	51.8	43.6		
		MCDB	40.7	43.2	53.4	67.4	76.9	80.2	83.2	83.0	75.8	65.7	54.4	44.4		
	5%	WB	36.8	38.7	44.6	54.8	63.8	67.9	70.0	68.6	62.2	56.2	48.9	41.0		
		MCDB	38.0	40.5	49.5	63.6	74.9	79.0	81.1	81.0	72.0	61.9	50.4	42.1		
	10%	WB	34.6	35.9	42.1	52.7	61.9	66.4	68.5	67.0	60.3	53.3	46.3	37.7		
		MCDB	35.5	37.1	46.6	61.1	71.9	76.8	79.0	78.3	69.1	58.1	47.7	38.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	11.2	14.7	20.0	21.9	21.4	21.6	22.7	20.4	17.3	9.9	8.8		
		MCDBR	8.8	12.5	22.1	25.9	26.6	26.4	26.5	29.7	28.5	24.5	13.9	10.3		
		MCWBR	7.7	9.7	13.7	13.2	11.9	11.2	10.0	11.3	13.2	15.0	10.8	9.0		
		MCDBR	8.3	11.6	19.7	22.1	23.0	22.8	22.7	24.4	23.9	21.7	12.4	10.1		
	5% WB	MCWBR	7.6	9.4	13.4	12.3	11.5	10.9	9.9	10.7	12.5	14.2	10.2	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.309	0.339	0.371	0.436	0.409	0.401	0.430	0.432	0.399	0.360	0.341	0.311		
	taud		2.216	2.163	2.221	2.121	2.280	2.323	2.252	2.232	2.288	2.376	2.315	2.262		
	Ebn at Noon		222	245	261	255	269	272	261	254	249	239	210	203		
	Edh at Noon		23	32	36	45	40	39	41	40	34	26	22	20		
(44) All-Sky Solar Radiation	RadAvg		275	537	944	1377	1802	1904	1849	1612	1099	631	252	187		
	RadStd		49	92	150	140	142	139	121	99	164	100	38	47		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.12	N/A	N/A	+1.76	+0.90	+1.08	N/A	-193	+200	+128
(47) Regional (10 neighbors)		+1.31	N/A	N/A	+1.80	+0.86	N/A	-252	-353	+214	+99

Nomenclature: See separate page